Tredence Analytics

JOB DESCRIPTION

Location:

Bengaluru, India

Role:

AI Engineering Intern (0–1 yr)

Duration:

6 months (extendable / convertible)

Stipend :

30k Per Month /-

About the Role

We are looking for an AI Engineering Intern who can work with our engineering team to build LLM-powered features, retrieval systems, and agent-style workflows. This role is ideal for someone strong in Python who wants hands-on exposure to modern AI engineering and backend systems.

Responsibilities

- Write clean, maintainable **Python** code for data processing, integrations, and automation.
- Build and maintain API endpoints using **FastAPI**, including async workflows where needed.
- Use **OpenAI and other LLM APIs** for generation, summarization, and extraction tasks.
- Build basic **RAG pipelines**: chunking, embedding, vector indexing, and retrieval.
- Work with vector DBs such as **FAISS, Chroma, or Milvus** for embedding search.
- Write SQL queries and work with relational databases like PostgreSQL/MySQL.
- Implement simple caching (in-memory, Redis) to optimize repeated requests.
- Use Python concepts such as **asyncio, threading, and multiprocessing** where applicable.

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- Participate in performance tuning by profiling Python code and optimizing bottlenecks.
- Contribute to documentation, testing, logging, and debugging of AI pipelines.
- Support small agent workflow prototypes and internal tool development.

Required Skills

- Strong command of **Python** and understanding of core libraries.
- Experience developing APIs with **FastAPI**.
- Familiarity with **async programming**, threading, and multiprocessing.
- Understanding of **LLM API usage** (OpenAI or similar).
- Knowledge of **RAG fundamentals** and experience with vector databases.
- Ability to write clear and efficient **SQL** for querying and schema design.
- Basic understanding of caching strategies (in-memory, Redis).
- Solid **problem-solving and DSA** fundamentals.
- Awareness of clean code practices and performance-minded programming.

Bonus / Nice-to-Have

- Exposure to **PyTorch** or other ML frameworks.
- Understanding of fine-tuning (LoRA or similar).
- Familiarity with **LangChain, LangGraph, or Google ADK**.
- Exposure to RLHF concepts.
- Experience with Kafka, RabbitMQ, Celery, or Airflow (even academic).
- Basic Docker or cloud familiarity.

Key Criteria

- **Builder Mindset**: You enjoy creating things end-to-end, iterating quickly, and figuring out practical solutions.

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- **Code Quality & Performance Awareness**: You care about writing efficient, reliable, and clean code.
- **Problem-Solving Ability**: Comfortable breaking down problems and applying DSA concepts when needed.
- **Learning Orientation**: Open to feedback, curious, and motivated to pick up new tools and workflows.

What You'll Gain

- Experience contributing to real AI systems across LLMs, RAG, and backend engineering.
- Hands-on exposure to modern AI infra (FastAPI, vector DBs, caching, async workflows).
- Mentorship from engineers working on production-grade AI systems.
- Opportunity for PPO or extension based on performance.